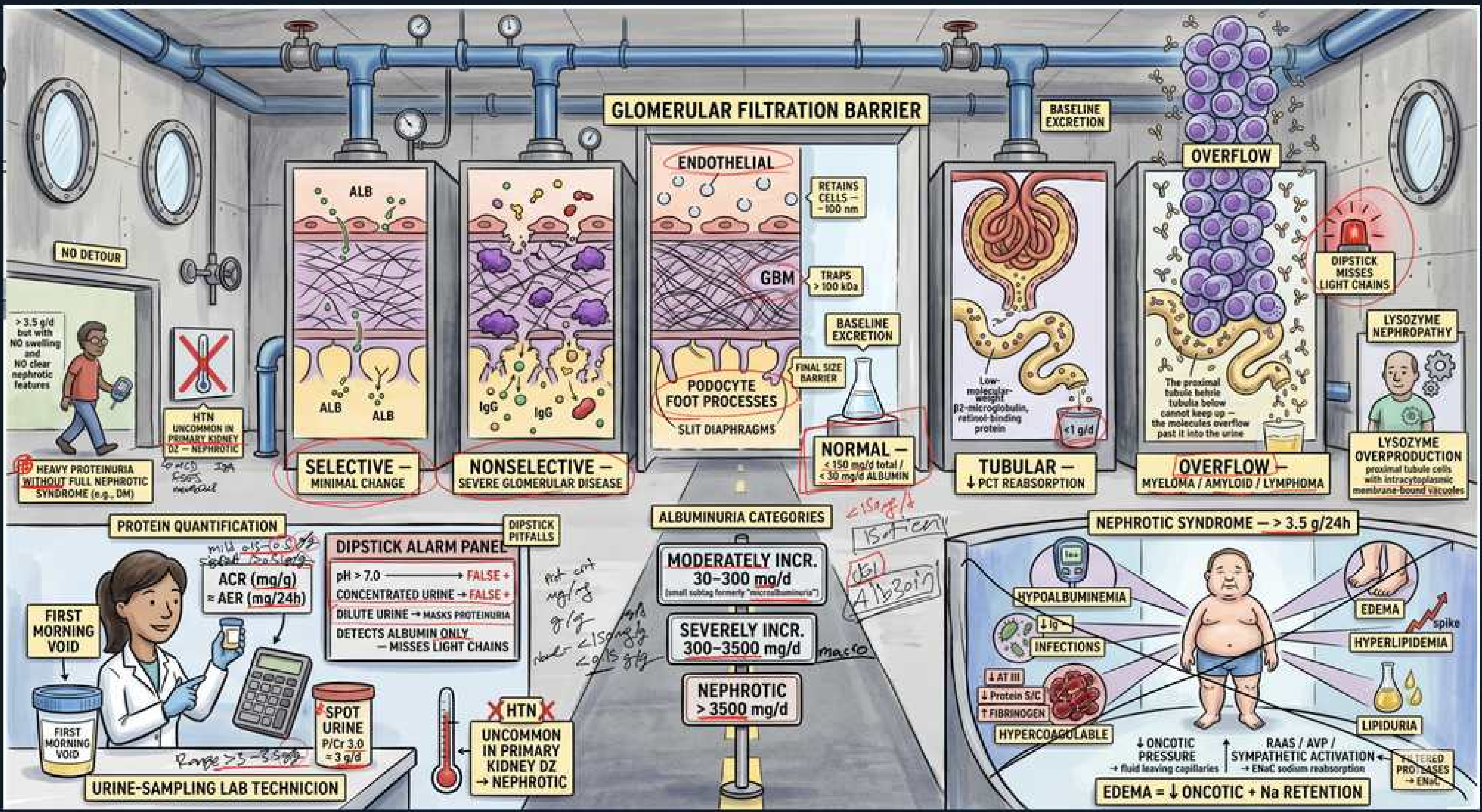
Proteinuria — Filtration Barrier, Categories & Nephrotic Syndrome



1. Glomerular filtration barrier

WHAT YOU SEE

Banner: glomerular filtration barrier

WHAT IT ENCODES

The barrier has three layers: fenestrated endothelium, glomerular basement membrane (charge/size barrier), and podocyte foot processes with slit diaphragms. Damage to any layer permits proteinuria. Albumin is normally repelled by negative charge.

BOARD TRAP

Wrong answer: tubular cells form the filtration barrier. Discriminator: the barrier is endothelium + GBM + podocyte slit diaphragm, not the tubule.

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2. Selective proteinuria — minimal change

WHAT YOU SEE

Selective panel, albumin only, minimal change

WHAT IT ENCODES

Selective proteinuria = mostly albumin lost, seen in minimal change disease where only podocyte foot-process effacement occurs (charge barrier lost, size barrier intact). Classic in children; steroid-responsive.

BOARD TRAP

Wrong answer: nonselective FSGS. Discriminator: selective (albumin-predominant) points to minimal change; nonselective points to more severe glomerular injury.

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3. Nonselective proteinuria — severe injury

WHAT YOU SEE

Nonselective panel, severe glomerular disease

WHAT IT ENCODES

Nonselective proteinuria includes larger proteins (albumin + globulins) and signals significant GBM/structural damage (e.g., FSGS, membranous, proliferative GN). Indicates loss of the size barrier.

BOARD TRAP

Wrong answer: minimal change disease. Discriminator: nonselective loss of large proteins indicates structural glomerular damage, not pure charge defect.

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4. Tubular proteinuria — failed reabsorption

WHAT YOU SEE

Tubular panel, low-MW proteins, PCT

WHAT IT ENCODES

Low-molecular-weight proteins (beta-2-microglobulin, lysozyme) are normally filtered then reabsorbed by the proximal tubule. Tubulointerstitial disease impairs reabsorption → tubular proteinuria (usually <2 g/day).

BOARD TRAP

Wrong answer: glomerular proteinuria. Discriminator: tubular proteinuria is low-MW (beta-2-microglobulin), modest amount, from PCT dysfunction not glomerular leak.

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5. Overflow proteinuria — light chains/myeloma

WHAT YOU SEE

Overflow panel, dipstick misses light chains

WHAT IT ENCODES

Overflow proteinuria: overproduction of filterable proteins overwhelms reabsorption (Bence-Jones light chains in myeloma, myoglobin, hemoglobin). Urine DIPSTICK detects albumin only and misses light chains — use urine protein electrophoresis.

BOARD TRAP

Wrong answer: negative dipstick rules out proteinuria. Discriminator: dipstick misses light chains — order urine immunofixation/SPEP/UPEP in suspected myeloma.

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6. Lysozyme nephropathy clue

WHAT YOU SEE

Lysozyme/monocytic leukemia note

WHAT IT ENCODES

Lysozymuria from monocytic/myelomonocytic leukemia is a cause of proximal tubular dysfunction and overflow low-MW proteinuria. A niche board association linking hematologic malignancy to renal protein handling.

BOARD TRAP

Wrong answer: glomerular nephrotic-range loss. Discriminator: lysozyme is a low-MW overflow protein from monocytic leukemia, not a glomerular leak.

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7. Protein quantification — ACR / AER

WHAT YOU SEE

Protein quantification, ACR mg/g, AER mg/24h

WHAT IT ENCODES

Quantify with albumin-creatinine ratio (mg/g, on a spot sample) or 24-hour albumin excretion rate. Spot ACR is preferred for screening and avoids timed-collection errors. First-morning void reduces orthostatic effect.

BOARD TRAP

Wrong answer: dipstick grading is sufficient for quantification. Discriminator: use spot ACR (mg/g) or 24h AER — dipstick is qualitative and concentration-dependent.

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8. Dipstick alarm panel — false negatives

WHAT YOU SEE

Dipstick: pH>7 false +, dilute urine false -

WHAT IT ENCODES

Dipstick pitfalls: very alkaline urine (pH>7) or highly concentrated urine gives false positives; dilute urine gives false negatives; and the strip detects ALBUMIN only, missing light chains. Confirm with ACR.

BOARD TRAP

Wrong answer: trust a negative dipstick. Discriminator: dipstick is concentration-dependent and albumin-only — confirm quantitatively and consider light chains.

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9. Albuminuria categories — A1/A2/A3

WHAT YOU SEE

Moderately incr 30-300, severely incr 300-3500

WHAT IT ENCODES

KDIGO ACR categories: A1 normal <30 mg/g; A2 moderately increased 30-300 (old microalbuminuria); A3 severely increased >300. Nephrotic-range proteinuria is >3500 mg/day. Albuminuria is an independent CKD/cardiovascular risk marker.

BOARD TRAP

Wrong answer: microalbuminuria is benign. Discriminator: A2 (30-300) signals early diabetic/CKD risk and predicts cardiovascular events.

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10. Nephrotic syndrome — >3.5 g/24h

WHAT YOU SEE

Nephrotic syndrome box, >3.5 g/24h

WHAT IT ENCODES

Nephrotic syndrome: proteinuria >3.5 g/day, hypoalbuminemia, edema, hyperlipidemia, and lipiduria. Reflects severe podocyte/GBM injury. Complications include infection and thrombosis.

BOARD TRAP

Wrong answer: nephritic syndrome. Discriminator: nephrotic = heavy proteinuria >3.5 g, bland sediment; nephritic = hematuria, RBC casts, hypertension, lower-grade proteinuria.

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11. Hypoalbuminemia → infections

WHAT YOU SEE

Hypoalbuminemia branch, infections

WHAT IT ENCODES

Urinary loss of immunoglobulins and complement factors (and antithrombin) predisposes nephrotic patients to encapsulated-organism infections like pneumococcal peritonitis. Hypoalbuminemia also drives edema.

BOARD TRAP

Wrong answer: infections are unrelated to nephrotic syndrome. Discriminator: loss of Ig/complement causes susceptibility to encapsulated organisms (e.g., pneumococcus).

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12. Hypercoagulable — antithrombin loss

WHAT YOU SEE

Hypercoagulable branch

WHAT IT ENCODES

Nephrotic syndrome causes a hypercoagulable state from urinary antithrombin III loss and increased hepatic procoagulant synthesis. Renal vein thrombosis is classic, especially in membranous nephropathy.

BOARD TRAP

Wrong answer: nephrotic patients bleed. Discriminator: nephrotic syndrome is PROTHROMBOTIC (antithrombin loss) — renal vein thrombosis, classically in membranous.

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13. Hyperlipidemia & lipiduria

WHAT YOU SEE

Hyperlipidemia / lipiduria branch

WHAT IT ENCODES

Hypoalbuminemia drives compensatory hepatic lipoprotein synthesis → hyperlipidemia; filtered lipids appear as lipiduria with oval fat bodies and Maltese-cross fatty casts under polarized light.

BOARD TRAP

Wrong answer: Maltese crosses indicate crystalluria/oxalate. Discriminator: Maltese-cross fatty casts under polarized light = nephrotic lipiduria.

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14. Edema = low oncotic + Na retention

WHAT YOU SEE

Edema bottom banner, oncotic + Na retention

WHAT IT ENCODES

Nephrotic edema results from reduced plasma oncotic pressure (hypoalbuminemia) plus primary renal sodium retention. Treatment combines loop diuretics, sodium restriction, and RAAS blockade to reduce proteinuria.

BOARD TRAP

Wrong answer: edema is purely from low oncotic pressure. Discriminator: primary renal Na retention also contributes — both mechanisms drive nephrotic edema.
